The Solutions for EE4.18 and AO6, 2020

Model answer to Q1(a): Bookwork and Discussions in Class

List five wireless applications that are currently being used within the security and defences industries and briefly describe (in a few sentences) how each application exploits the radio frequency spectrum.

The student can state any 5 of the following (and/or can give new examples) and include/expand on what has been written below:

Phased-array radar – narrow beams can be created when the effective size of the antenna is large, so long as the elements are spaced by half a wavelength. Therefore, large antennas having 1000s of elements are generally only practical at microwave-wave frequencies or above.

Electronic warfare – examples include electronic surveillance measures, electronic counter-measures, electronic counter-counter-measures and decoys. These application can operate across the whole spectrum.

Smart munitions – a millimetre-wave radar/camera can be located at the front of a missile, and used as a proximity fuse or to trigger detonation.

Secure Communications – these application can operate across the whole spectrum.

Explosive/Biological weapon detection – microwave/millimetre-wave/THz spectroscopy can be used to detect spectral absorption lines of unknown samples.

Active denial – used for crowd control in the US. People are irradiated by millimetre-wave radiation from a very high gain parabolic antenna, which provides very high directivity for selecting individuals.

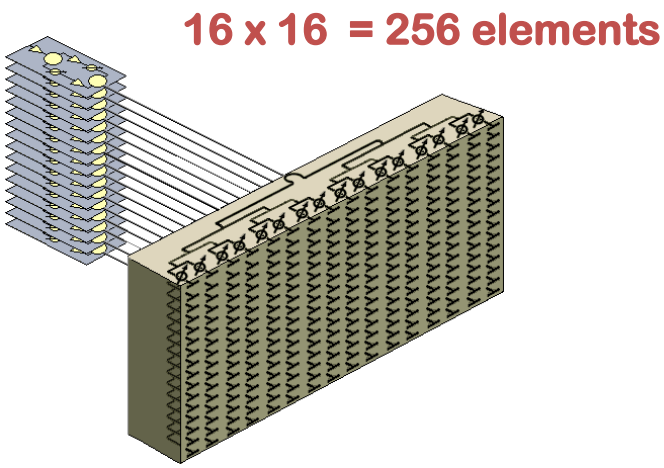
Remote sensing, for surveillance – examples include microwave and millimetre-wave radiometric imaging, now being deployed in airports across the world. The spatial contrast in thermal noise generated by an individual can be detected and attributed to various materials (e.g., metals, liquids and powders), which appear out of place under normal conditions.

[10]

Model answer to Q1(b): Bookwork and Discussions in Class

State a well-known example of how an older technology has been updated by adopting RF electronics and explain how this change provides an adaptive and agile solution for modern warfare.





Answers

Phased-array antennas, used in modern military radars, is one of the best known examples. The old and very large parabolic antenna is mechanically very slow and can cause stability problems (due to inertia) when located at the nosecone of a high-speed fighter aircraft. Moreover, it can only track one target at a time. In contrast, a fixed large-element phased array radar can track many independent fast-moving targets at a time. Behind each element there could be a transmit-receive module with programmable phase shifter and programmable attenuator, which act as vector modulators.

[4]

Model answer to Q1(c): Bookwork and Discussions in Class

From a technical perspective, explain how RF electronics can be used for crowd control.

Active Denial System (or Heat Ray) – This is a non-lethal, directed-energy weapon developed by the U.S. military and police. It is a high power millimetre-wave transmitter, primarily used for crowd control, which can operate up to a distance of 1 km. The millimetre waves excite the water and fat molecules on the skin, instantly heating it and causing intense pain.



[3]

Model answer to Q1(d): Bookwork and Discussions in Class

From a technical perspective, explain the limitation of autonomous cruise control.

This technology was first introduced back in the 1970s as a "car radar". It operates at 76.5 GHz in Europe, to give a reasonably good angular resolution (~2°) from a high gain antenna. It can "see" up to a range of ~200 m with 0.5 m resolution. Unfortunately, the radar has difficulties operating in fog/rain/snow, due to high atmospheric attenuation. Moreover, undesirable off the road backscatter can confuse the radar when it is turning corners. As a result, today, these systems are only recommended for when the car is cruising along straight roads. Currently, advances are being made to include data fusion techniques with acoustic, infrared and visual spectrum sensors, in order to make the radar less prone to false detection.

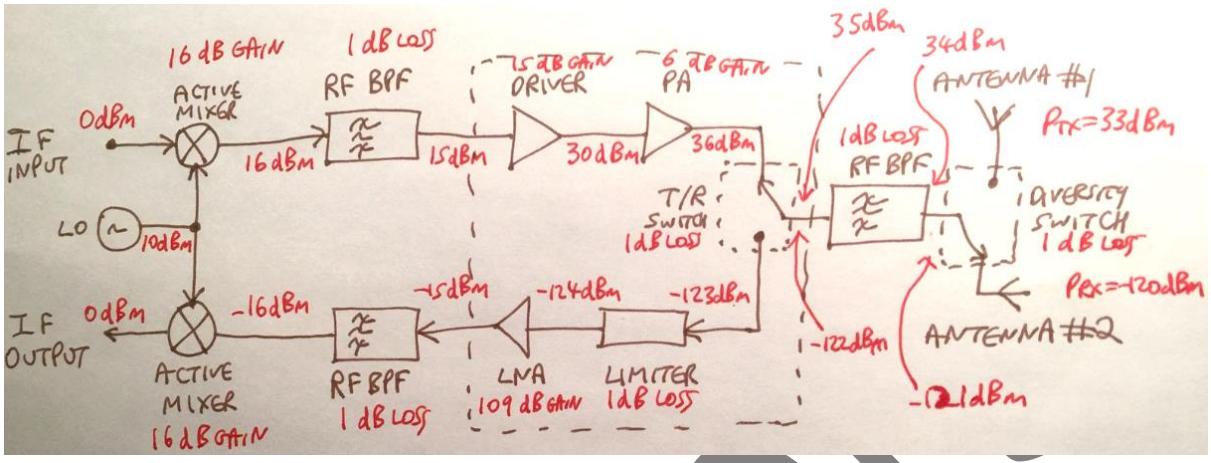


[3]

Answers

Model answer to Q 2(a): Bookwork and Discussions in Class and Computed Example

Draw the generic block diagram of a half-duplex transceiver, between IF and RF ports, indicating the directions of signal flow. Clearly identify each block and assign a reasonable value for the power gain or loss typically found for each. Then determine typical power levels at the input and output of each block. All assumptions must be clearly stated; for example, typical input and output power levels.



Assumptions are that typical input and output power values are used that correspond to a GSM mobile phone (the student can choose their own values, but these should be justified with a realistic scenario): P_{IF} = 0 dBm; P_{TX} = 33 dBm; P_{RX} = -120 dBm

All mixers are active and have a conversion gain of 16 dB, typically fed by a LO with a pump power of 10 dBm. The filters, switches and limiter all have a 1 dB insertion loss. The amplifier gains can then be calculated.

[10]

Model answer to Q 2(b): Bookwork and Discussions in Class

i) Briefly explain the problem with both short-term and long-term stability associated with the LO.

Short-term stability corresponds largely to the injection of phase noise into the mixer, which reduces the signal/noise ratio at the output. Long-term stability results in frequency drifting and attenuation of the signal getting into the following stage.

[2]

ii) Briefly explain the problem of LO-RF leakage when operating in 'radio silence'.

Poor LO-RF leakage means that a very small fraction of the high (e.g. 10 dBm) LO power could escape through the receive chain and be detected by a highly sensitive location finding receiver, which when triangulated could give away the location of the transceiver when operating in Radio Silence.

[2]

iii) Why does the PA need to be driven? Briefly compare and contrast the gain and output power in both stages.

The PA is primarily designed for optimal efficiency, located near the maximum output power level, and so its power gain is generally a secondary issue and usually low. In contrast, the driver is primarily designed for optimal power gain for the specified low output power needed at the input of the PA.

[2]

iv) Briefly explain the problem with poor isolation in the T/R Switch and estimate a typical poor isolation level.

Answers

With poor isolation, the high transmit power (e.g. 36 dBm or 4 W) can leak through the T/R switch and saturate (or even destroy when unprotected) the LNA that may be expecting signal levels as low as -124 dBm. Therefore, in theory, T/R switches that operate in a single-channel half-duplex system need an isolation of 160 dB. In practice, 2-channel systems (e.g. GSM) can achieve the required level of isolation by including some additional channel filtering.

[2]

v) What is the function of a Diversity Switch?

The Diversity Switch is used for switching between 2 or more antennas, to improve transmission-reception. For example, two antennas could have orthogonal polarizations or be just spatially separated, it may prove beneficial to switch between the two when transceiver locations change.

[1]

vi) What is the function of a Limiter?

The limiter is simply a power limited (e.g. back-to-back diodes), placed at the input of the sensitive LNA, which clamps excess input signal voltage levels to that of the built-in barrier potential of the protection diodes.

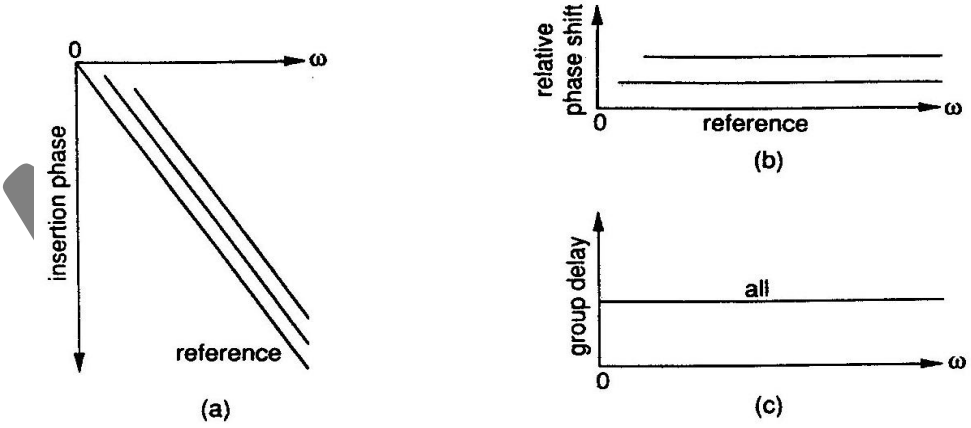
[1]

Model answer to Q 3(a): Bookwork

Mathematically define the characteristics of a true phase shifter and illustrate the frequency responses of relevant performance parameters $\Delta\theta(\omega)$ and $\Delta\tau(\omega)$.

Relative Phase Shift, $\Delta\theta(\omega) = \Delta\theta(\omega_o)$

Differential-phase group delay variation, $\Delta\tau(\omega) = 0$.



[4]

Model answer to Q 3(b): New Derivation

If the antenna in Figure 3.1 employs true phase steering, mathematically prove that the direction of the main beam from broadside ∅ is a function of frequency. What is this phenomenon called?

$$\sin\emptyset = \frac{\Delta\theta(\omega)}{\theta(\omega)} \quad \text{where} \quad \theta(\omega) = \frac{2\pi}{\lambda}\cdot d = \frac{\omega}{c}\cdot d \quad \text{and so} \quad \sin\emptyset = \frac{\Delta\theta(\omega)}{\omega\tau} \quad \text{giving} \quad \emptyset = \sin^{-1}\left(\frac{\Delta\theta(\omega)}{\omega\tau}\right)$$

$$True\ Phase\ Steering\text{:}\ \Delta\theta(\omega) = \Delta\theta(\omega_o) \quad \therefore \emptyset = \sin^{-1}\left(\frac{\Delta\theta(\omega_o)}{\omega\tau}\right) = f(\omega)$$

This phenomenon is called 'phase squinting'.

[4]

Model answer to Q 3(c): Discussion in Class

What physical separation distance between elements *d* is commonly used and why? Also, what practical limitation does this pose when scaling with frequency?

$$\text{To minimize unwanted sidelobes: } d \to \lambda_o/2 \quad \text{and so} \quad \theta(\omega) \to \pi \frac{\omega}{\omega_o}$$

As the operating frequency increases, it may be difficult to contain all the RF electronics behind each element, especially with a radar or communications system that needs separate T/R modules.

[4]

Model answer to Q 3(d): New Derivation

With the separation distance state in 3(c), derive the resulting expression for $\emptyset(\omega)$.

$$\emptyset \to \sin^{-1}\left(\frac{\Delta\theta(\omega)}{\pi} \cdot \frac{\omega_o}{\omega}\right) \Longrightarrow \sin^{-1}\left(\frac{\Delta\theta(\omega_o)}{\pi} \cdot \frac{\omega_o}{\omega}\right) = f(\omega)$$

[3]

Model answer to Q 3(e): New Application of Theory

Using the expression derived in 3(d), calculate $\emptyset(\omega)$ for a 90° relative phase shift for the following three values of $\omega/\omega_o = [0.9, 1.0, 1.1]$.

$$\emptyset(\omega/\omega_o) = [26.7°, \quad 30.0°, \quad 33.4°]$$

[3]

Model answer to Q 3(f): New Application of Theory

If the scenario in 3(e) is implemented by a 4-element phased array antenna, what is the maximum relative phase shift that is needed?

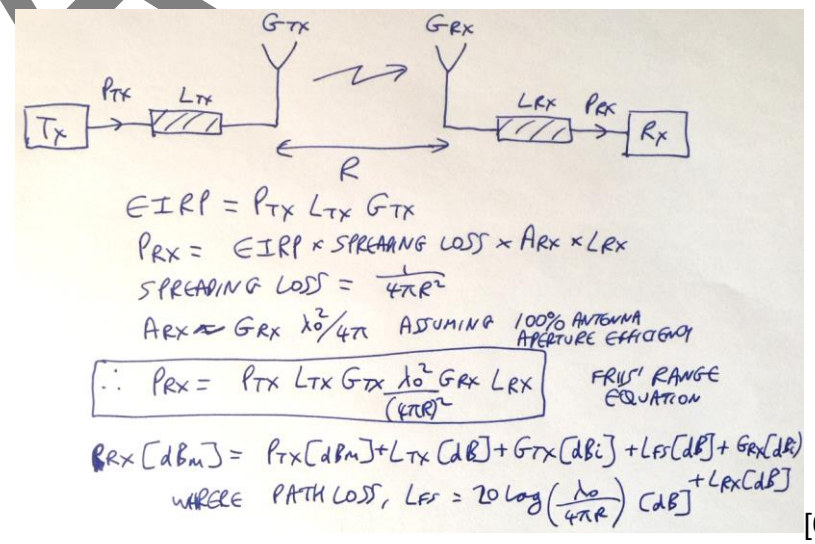
The various elements would need a relative phase shift distribution of 0°, 90°, 180° and 270°.

[2]

Model answer to Q 4(a): Extended Bookwork and Discussions in Class

Consider a wireless line-of-sight communications link operating at 76.5 GHz.

With the use of a simple sketch (clearly labelled with the appropriate variables), derive Friis' link equation in log form (showing the appropriate units for each variable) for a line-of-sight wireless link. Interconnecting feed line losses must be included.

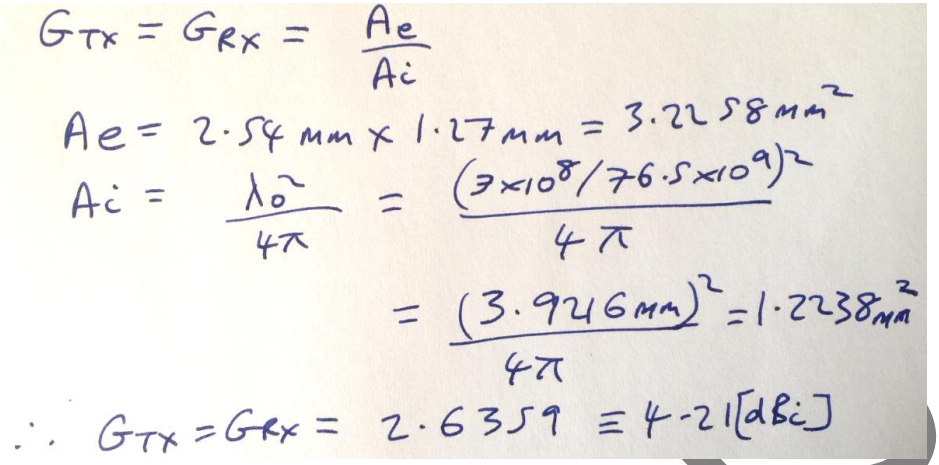


[6]

Answers

Model answer to Q 4(b): New Calculation

Tx and Rx antennas are identical and made from the open-ended apertures of the associated WR-10 waveguide feed lines, having internal cross-sectional dimensions of 2.54 mm x 1.27 mm. Assuming a 100% antenna efficiency, calculate the effective antenna gain for these apertures.

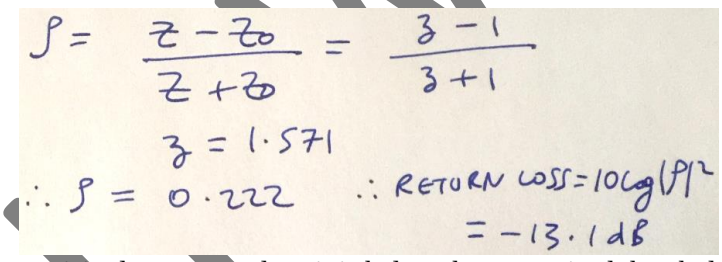


[2]

Model answer to Q 4(c): Discussions in Class and New Derivation

The feed lines have a wave impedance normalized to that of free space of 1.5716. Calculate:

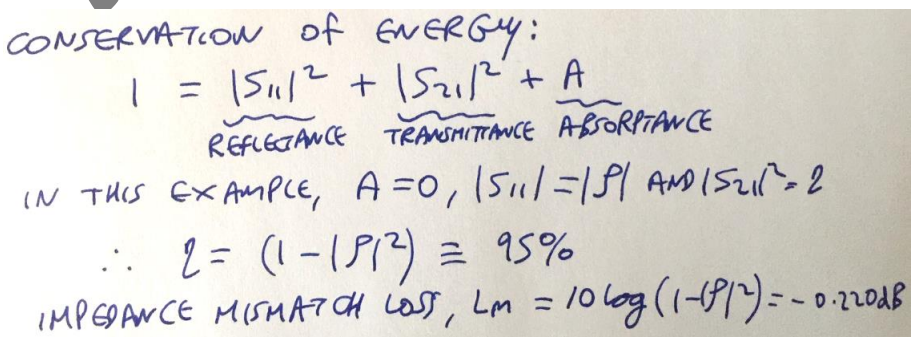
i) The voltage-wave reflection coefficient at the aperture (boundary between the waveguide and free space) and the resulting return loss. Comment on the significance or otherwise of this return loss.



The return loss is surprisingly very good, as it is below the recognised threshold of -10 dB for an impedance matched device.

[2]

ii) Mathematically, define the principle of 'Conservation of Energy', in relation to the S-parameters for a symmetrical and reciprocal passive 2-port network, and then apply this to calculate the antenna efficiency and the impedance mismatch loss at the aperture.



[4]

AI-NOT

Model answer to Q 4(d): New Calculation

Calculate the minimum combined far-field distance between the Tx and Rx antennas in 4(b).

$$R_{ff} = \frac{2(L_{TX}^2 + L_{RX}^2)}{\lambda_0} = \frac{4 L_{TX}^2}{\lambda_0}$$

$$L_{TX} = L_{RX} = \sqrt{2.54^2 + 1.27^2} \text{ [mm]}$$

$$\therefore = 2.8398 \text{ mm}$$

$$\therefore R_{ff} = 8.226 \text{ mm}$$

[2]

Model answer to Q 4(e): New Calculation

The Tx power is 30 mW and the identical interconnecting feed lines have an insertion loss of -0.1 dB that are connected to the identical aperture antennas (which are perfectly aligned for optimal line-of-sight communications). Choose any value of range within the far field, between the antennas, and calculate:

i) Path loss.

FREE-SPACE PATH LOSS

$$L_{FS} = \left(\frac{\lambda_0}{4\pi R}\right)^2 = 9.739 \times 10^{-4} \equiv -30.1 \text{ dB}$$

[2]

ii) Rx power.

A range of 10 mm was chosen, as this is greater than the far field distance of 8.266 mm.

$P_{TX} = +14.77 \text{ dBm} \equiv 30 \text{ mW}$

$L_{TX} = -0.10 \text{ dB}$

$G_{TX} = +4.21 \text{ dBi}$

$L_{MTX} = -0.22 \text{ dB}$

$L_{FS} = -30.10 \text{ dB}$

$G_{RX} = +4.21 \text{ dBi}$

$L_{MRX} = -0.22 \text{ dB}$

$L_{RX} = -0.10 \text{ dB}$

$P_{RX} = -7.55 \text{ dBm} \equiv 175.8 \mu\text{W}$

[2]

Answers

Model answer to Q 5(a): New Application of Discussions in Class

Consider the three-port broadband directional coupler shown in Figure 5.1. Holes spaced $\lambda_g/4$ apart allow waves travelling towards Port 3 to combine, and ideally cancel in the opposite direction. A manufacturer has given the following relevant specifications:

Coupling:	10 dB	Directivity (or Isolation):	40 dB
Main Line VSWR:	1.10	Auxiliary Line VSWR:	1.17

Stating all assumptions, extract the complex S-parameter matrix $[S^c]$ for this coupler, given in equation 5.1, for this 3-port network using the given manufacturer's specifications.

$$[S^c] = \begin{pmatrix} S_{11}^c & S_{12}^c & S_{13}^c \\ S_{21}^c & S_{22}^c & S_{23}^c \\ S_{31}^c & S_{32}^c & S_{33}^c \end{pmatrix} \quad (5.1)$$

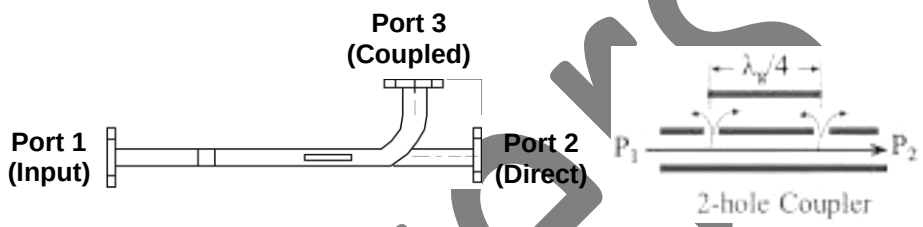


Figure 5.1 Three-port Broadband Directional Coupler.

COUPLING = 10dB

$\therefore |S_{31}| = |S_{13}| = 0.316$

$\angle S_{31} = \angle S_{13} = -90°$ ASSUMED

$\rho = \frac{VSWR-1}{VSWR+1}$

$\therefore |S_{11}| = |S_{22}| = 0.048$

$\angle S_{11} = \angle S_{22} = 0$ ASSUMED

$|S_{33}| = 0.078$

$\angle S_{33} = 0$ ASSUMED

CONSERVATION OF ENERGY GIVES

$1 = |S_{21}|^2 + |S_{31}|^2$ FOR A LOSSLESS NETWORK (ASSUMED)

$\therefore |S_{21}|^2 = |S_{12}| = \sqrt{1-|S_{31}|^2} = 0.949$

$\angle S_{21} = \angle S_{12} = -90°$ SINCE $\angle S_{31} = \angle S_{13} = -90°$

DIRECTIVITY = 40dB $\therefore |S_{32}| = |S_{23}| = 0.01$ AND $\angle S_{32} = \angle S_{23} = 0$ ASSUMED

$$[S^c] = \begin{pmatrix} 0.048 & -j0.949 & -j0.316 \\ -j0.949 & 0.048 & 0.01 \\ -j0.316 & 0.01 & 0.078 \end{pmatrix}$$

[5]

Model answer to Q 5(b): New Application of Discussions in Class

A four-port reflectometer can be used to measure the Return Loss of a Device Under Test (DUT). Connecting power detectors to Ports 2 and 4 of the four-port network shown in Figure 5.2 is one example. Using intuitive reasoning, briefly explain the operation of this circuit for measuring a one-port DUT that presents an arbitrary impedance to Port 3. Briefly explain what happens to the measured Return Loss if too much RF power is generated by the source.

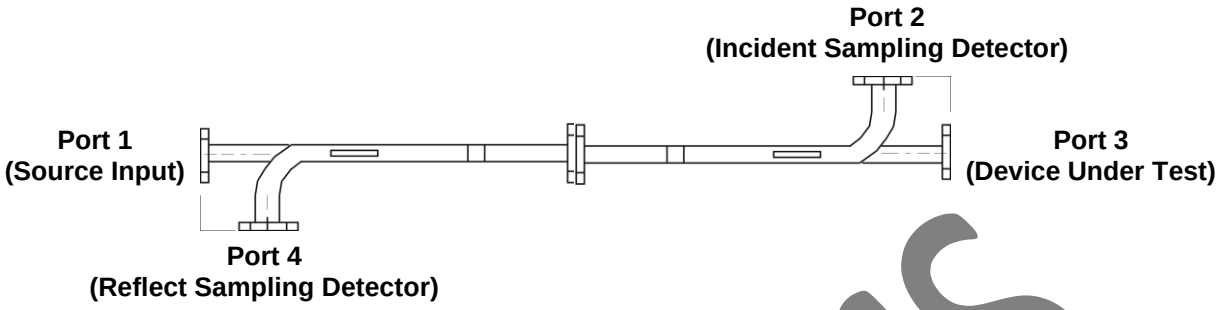
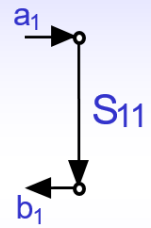


Figure 5.2 Four-port Reflectometer (source, power detectors and DUT not shown).

A small-signal sinusoidal signal generator is connected to Port 1; a square-law power detector is connected to Port 2, for sampling the power incident on the DUT; an identical square-law power detector is connected to Port 4, for sampling the power reflected from the DUT; and the DUT is connected to Port 3.

S-parameters treat the signal in terms of voltage waves (actually square root of power for a reference impedance - the "system impedance", which is normally 50 Ω)



Nodal Notation (2-nodes per port)
a_1: voltage wave coming into port1
b_1: voltage wave coming out from port1

Branch Notation (signal flow paths)
S_{11}: Reflection back from the input (INPUT MATCHING)

This leads to the definitions of the S-parameters:

$$S_{11} = b_1/a_1$$

With square-law detectors, the output voltage from the detectors at Port 2 is proportional a_1^2, while that from the detector at Port 4 is proportional to b_1^2. In principle, with calibration, it is then possible to extract the magnitude of the voltage-wave reflection coefficient:

$$Input\ Return\ Loss,\ RL_{IN} = 10\log|S_{11}|^2$$

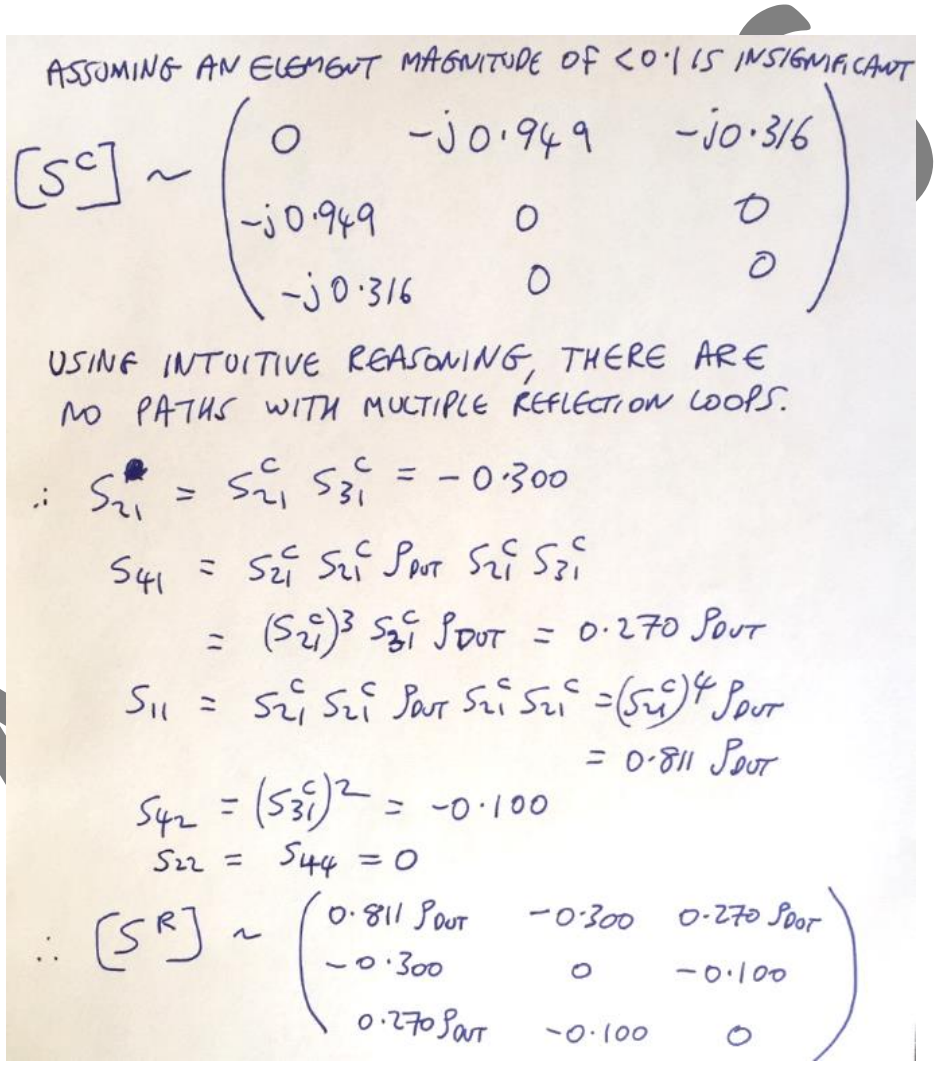
If the RF power level from the source is too high then the power detectors will no longer exhibit the square-law response and so the return loss will no longer be represent an accurate measurement.

[5]

Model answer to Q 5(c): New Application of Discussions in Class

Now apply reasonable simplifying approximations to the S-parameter matrix in 5(a). Hint: if the magnitude of a matrix element is less than 0.1 then it can be set to zero. Using the simplified $[S^c]$ elements and intuitive reasoning, determine the complex S-parameter matrix $[S^R]$, given in equation 5.2, for the four-port reflectometer network with Port 3 terminated by a one-port DUT having an arbitrary voltage-wave reflection coefficient ρ_{DUT}. The source and power detectors can be assumed to have perfect impedance matching.

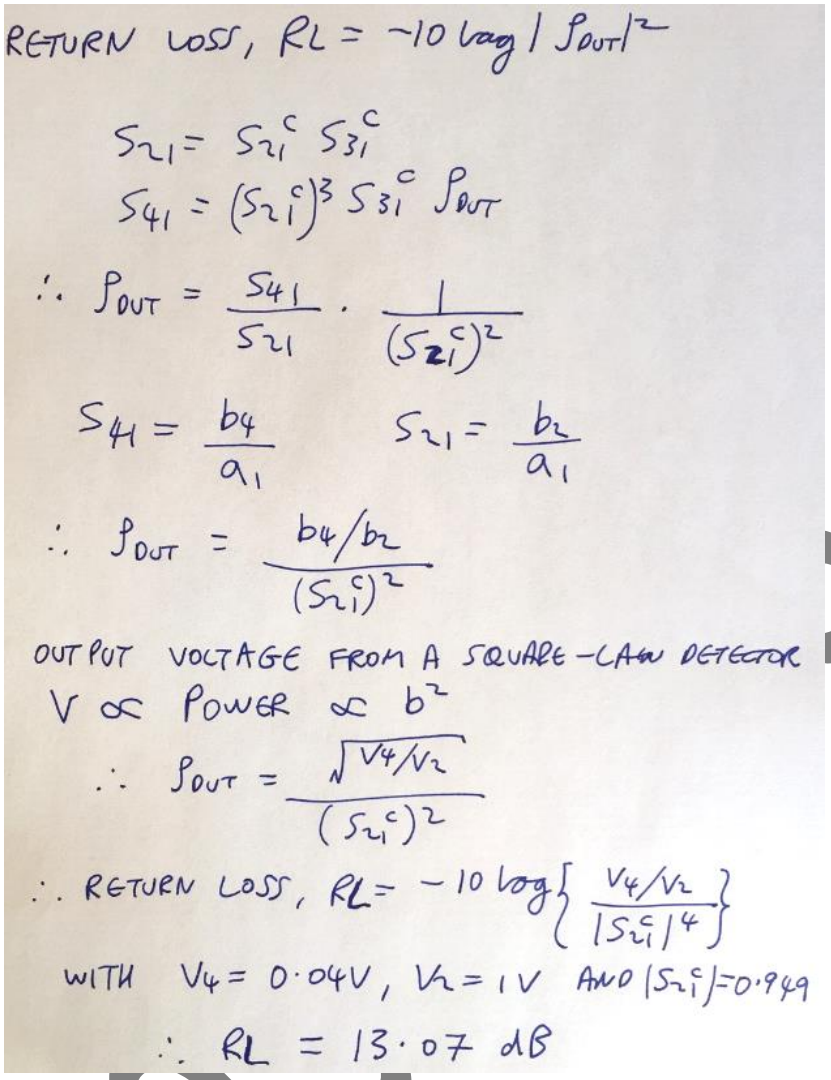
$$[S^R] = \begin{pmatrix} S_{11} & S_{12} & S_{14} \\ S_{21} & S_{22} & S_{24} \\ S_{41} & S_{42} & S_{44} \end{pmatrix} \quad (5.2)$$



[5]

Model answer to Q 5(d): New Application of Discussions in Class

Assume that $S_{21} \approx b_2/a_1$ and $S_{41} \approx b_4/a_1$ in 5(c), where a and b respectively represent voltage waves entering and leaving the network port, and square-law detectors are employed (operating in their linear region) producing corresponding output voltages V_2 and V_4. Determine an expression for the measured Return Loss of the Device Under Test in terms of V_2, V_4 and S_{21}^c. Using S_{21}^c from 5(a), and measured values of V_4 = 0.04 V and V_2 = 1 V, calculate the measured Return Loss.



[5]

Model answer to Q 6(a): Bookwork and Computed Calculation

Design a 3rd order Chebyshev *LC* band-pass filter to meet the following specifications:

Centre frequency of operation: $f_o = 2$ GHz
-3 dB Bandwidth: 20 MHz
Worst-case passband ripple: 0.1 dB
Source and load impedances: $Z_o = 50\ \Omega$
First branch: Series

Design the filter with the use of series and shunt *LC* resonators.

From filter tables provided: *g1* = 1.433; *g2* = 1.594; *g3* = 1.433

- With low-pass filters, de-normalising is performed directly:

series $L_k = \frac{g_k Z_L}{\omega_c}$ $C_k = \frac{g_k}{\omega_c Z_L}$ shunt

- With high-pass filters, de-normalising is performed by taking the dual of the LPF:

series $C_k = \frac{1}{g_k \omega_c Z_L}$ $L_k = \frac{Z_L}{g_k \omega_c}$ shunt

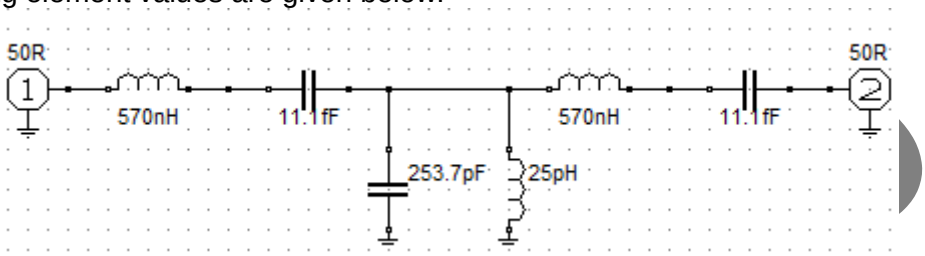
- With band-pass filters, replace the series element by a series tuned circuit and the shunt element by a parallel tuned circuit, all with $g_k \rightarrow g_k/\Delta$ and $\omega_c \rightarrow \omega_o$:

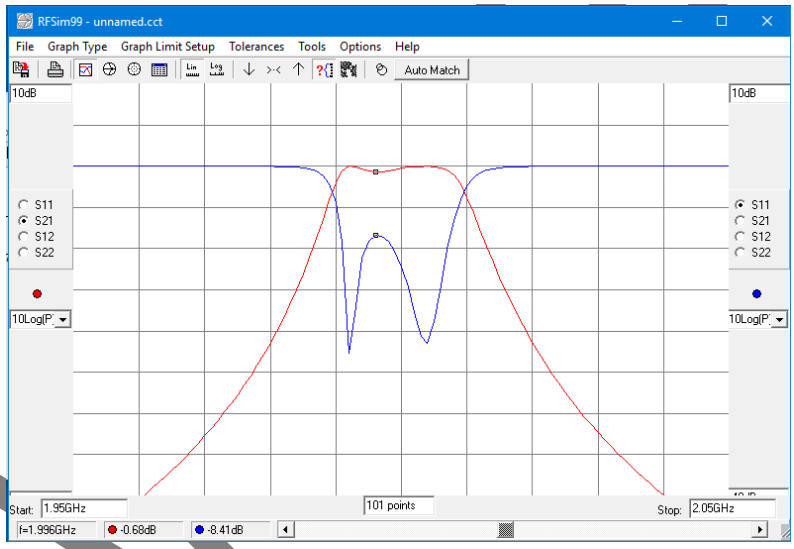
$\Delta = \frac{\omega_2 - \omega_1}{\omega_o} \equiv \frac{1}{Q_L}$ fractional -3 dB bandwidth

with respective upper and lower -3 dB cut-off frequencies ω_2 and ω_1

$\omega_o = \sqrt{\omega_1 \omega_2}$ *geometric* mean angular frequency

The resulting element values are given below:

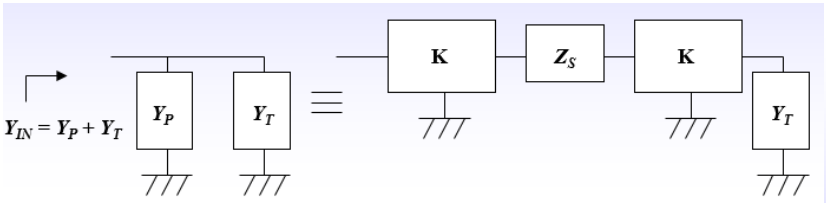


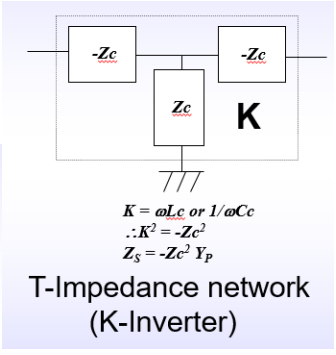


[10]

Model answer to Q 6(b): Bookwork and New Calculation

Redesign the filter in 6(a) to avoid the use of shunt *LC* resonators by employing inductor-based K-inverters having $K^2 = 1{,}000$.

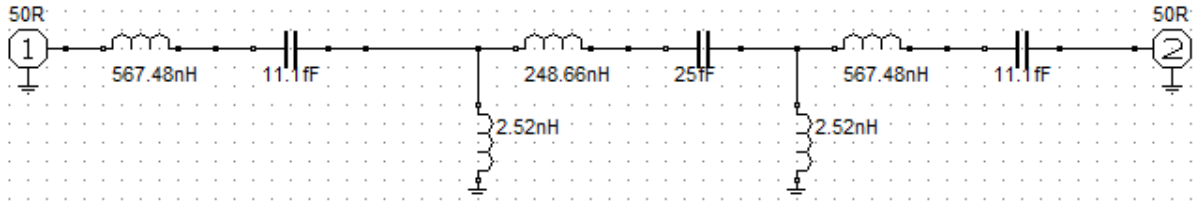


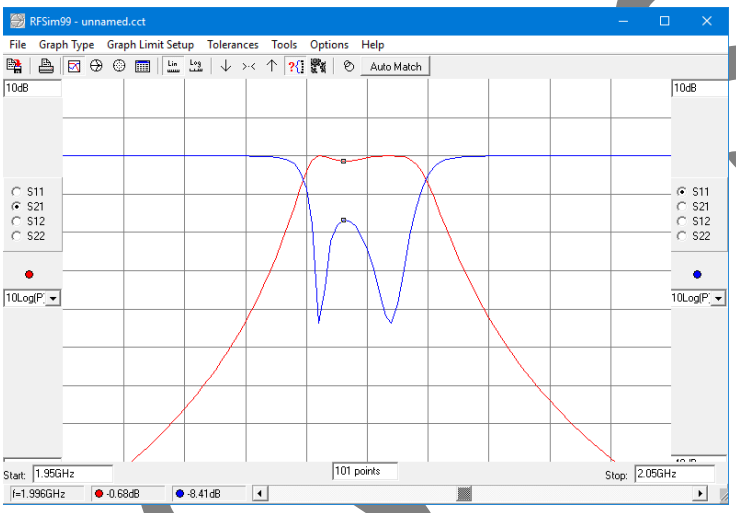


$$K = \omega Lc \qquad Lc = 2.52\ \text{nH} \qquad Z = K^2\, Y$$

$$Ls = K^2 Cp = 253.7\ \text{nH} \qquad Cs = Lp/K^2 = 25\ \text{fF}$$

The resulting element values are given below:





[10]

Answers